DHRUV ASHOK PATEL

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Cybersecurity engineering graduate student building detection and response systems across cloud and endpoint, including an AWS pipeline that auto-reverts risky changes in seconds, ATT&CK-mapped Wazuh and Splunk detections, malware reverse engineering, and LLM prompt injection research. Sharpening skills daily through bug bounty hunting, CTFs, and hands-on labs.

Education

Master of Science in Cybersecurity Engineering | University of Southern California Aug 2025 – May 2027
GPA: 3.7/4.0 | **Coursework:** Computer Forensics (DSCI 528); Malware Analysis & Reverse Engineering (TAC 443); Security & Privacy (DSCI 529); Computer System Assurance (DSCI 523); Security Systems (CSCI 530)

B.E. Computer Science & Engineering (Cybersecurity) | University of Mumbai Dec 2021 – May 2025
GPA: 9.1/10 | **Coursework:** Digital Forensics, Fundamentals of Information Security, Cybersecurity Risk Management, Ethical Hacking & Cybersecurity, ML for Cybersecurity

Projects

AWS Detection & Auto-Remediation Pipeline | Python, Terraform, AWS GitHub

- Engineered an event-driven AWS security control (CloudTrail, EventBridge, SQS, Lambda, DynamoDB, SNS) that auto-reverts public S3 buckets, IAM privilege escalation, and exposed EC2 security groups in a median 2.9 to 9.5 s across 30 timed runs.
- Mapped detections to MITRE ATT&CK (T1530, T1098, T1562.007), cut false positives with tag-based allowlists and a dry-run mode, and routed failed events to a dead-letter queue; validated with 60 pytest unit tests and attack simulations.
- Deployed via Terraform under least-privilege, remove-only IAM and blocked forged events with an explicit-deny SQS policy.

SOC Detection Engineering Lab | Wazuh, Sysmon, Splunk, Python GitHub

- Simulated 5 attack scenarios against a Windows 11 endpoint and developed 15+ Wazuh detections covering 8 MITRE ATT&CK techniques across credential access, defense evasion, persistence, and execution.
- Automated alert triage with VirusTotal indicator enrichment and forwarded 500+ alerts into Splunk for SPL threat hunting.

LLM Agent Security: Indirect Prompt Injection Lab | Python, Anthropic API, React GitHub

- Threat-modeled and implemented 5 indirect prompt injection attack classes (data exfiltration, multi-turn manipulation, tool abuse, RAG poisoning, agent-to-agent cascading) against a tool-using LLM agent.
- Designed 5 defenses and an evaluation harness scoring all 25 attack and defense pairs, with a React interface for live detections.

Cloud Threat Detection Lab | GCP Cloud Audit Logs, Splunk, Python GitHub

- Modeled a GCP IAM privilege escalation (roles/viewer to roles/owner via SetIamPolicy), MITRE ATT&CK T1078.004.
- Built a Python pipeline analyzing 32+ audit events and forwarding them to Splunk, with SPL detections and an hourly alert.

Malware Reverse Engineering: BRBBot Analysis | x64dbg, scdbg, blobrunner, CyberChef TAC 443 | 2026

- Deobfuscated a remote access trojan's PowerShell loader, extracted its shellcode, and resolved ROR13 API hashing.
- Identified command-and-control servers, mapped behavior to ATT&CK, and documented indicators of compromise.

Quantum-Secured Payment System | Python, FastAPI, React, MongoDB, Docker GitHub

- Built a payment platform with BB84 quantum key distribution, eavesdropper detection, and AES-256-GCM encryption.
- Reviewed the codebase and fixed hardcoded keys, weak random number generation, and AES misused for passwords.

Experience

Cybersecurity Intern | Crossover Technology Pvt. Ltd. Mar 2025 – Jun 2025

- Triaged 50+ SIEM alerts daily, escalating confirmed threats by severity and improving triage efficiency by 30%.
- Automated parsing of Nessus scan output with Python scripts and prioritized findings by CVSS score for remediation.
- Investigated anomalous activity in system logs and documented findings in reports and incident response runbooks.

Skills

Detection and Response: Splunk (SPL), Wazuh, Sigma, Sysmon, MITRE ATT&CK, Nessus, Wireshark, Nmap, Burp Suite
Malware Analysis and Forensics: x64dbg, Ghidra, scdbg, CyberChef, Autopsy, EnCase, FTK, static and dynamic analysis
Languages and Systems: Python, Bash, PowerShell, C/C++, SQL, JavaScript, Linux, Windows, Active Directory
Cloud & IaC: AWS (IAM, CloudTrail, EventBridge, Lambda, SQS, S3, EC2, DynamoDB, SNS), GCP, Terraform
AI/LLM Security and Cryptography: Prompt injection, LLM agent tool abuse, MITRE ATLAS, AES-256-GCM

Publications

Advancing Security in Digital Transactions using Quantum Cryptography, Journal of Information Systems Engineering and Management, Apr 2025
Malware Classification using Deep Learning, International Research Journal of Engineering and Technology, Oct 2024

Activities, Certifications and Awards

- **Bug Bounty and CTFs:** Hunting on HackerOne AI/LLM programs; 50+ TryHackMe and HackTheBox labs; CTF competitions
- **Certifications and Awards:** Google Cybersecurity Specialization | 1st Place, WordHack Competition (CyberPeace Foundation)